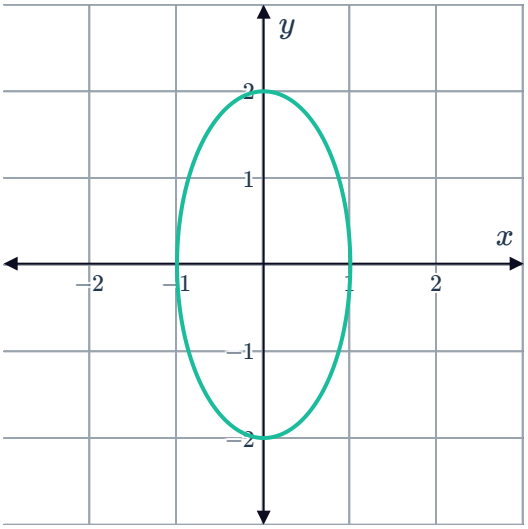
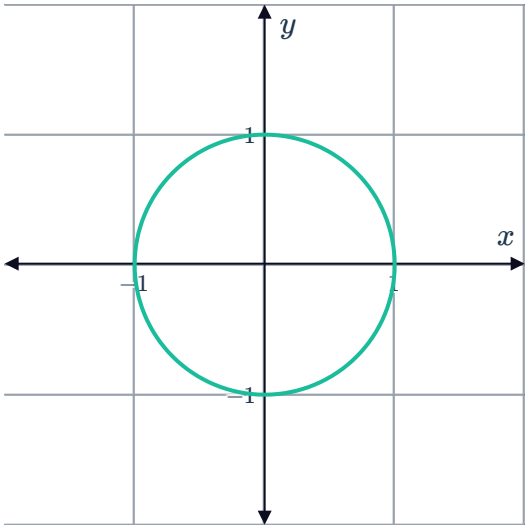
Equations of circles

1 Consider all points that are equidistant from the origin. Which of following graphs represent this collection of points?

Graph A



Graph B



2

a Plot four points such that they are all exactly 4 units away from the origin.

b What shape is formed by connecting these points with four straight lines?

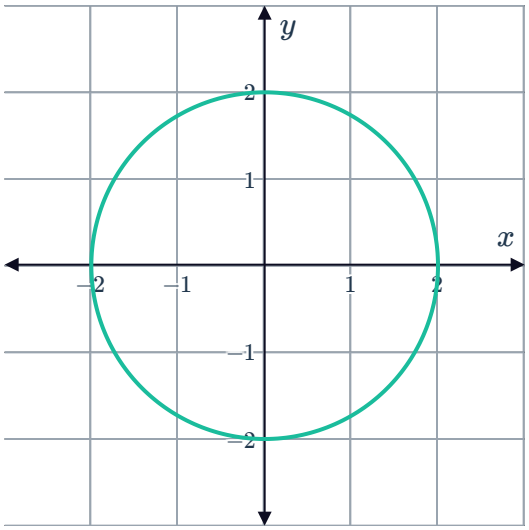
c As the number of points we plot and connect gets larger and larger, what shape will be formed?

3 For each of the following circles, state:

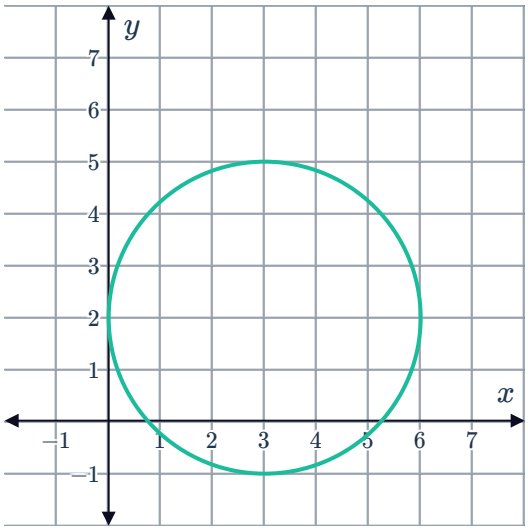
i The coordinates of the center.

ii The radius.

a

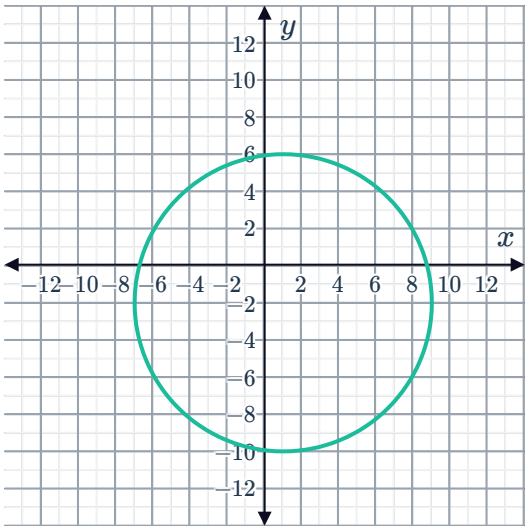


b

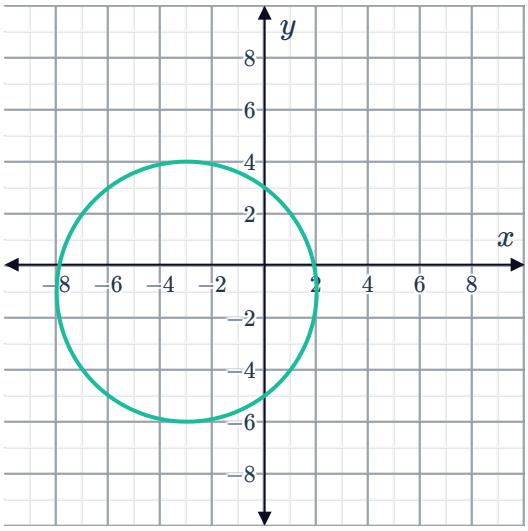


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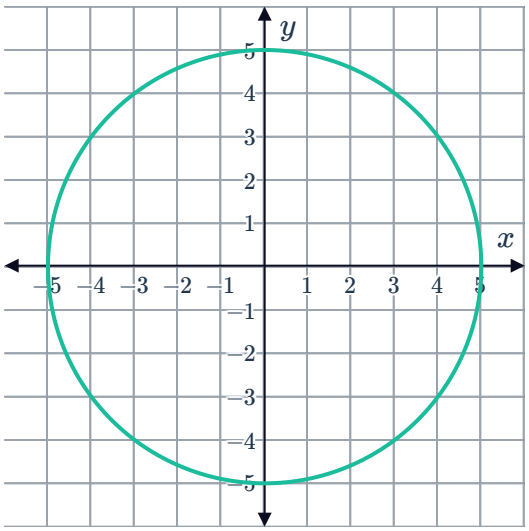
c



d



4 Consider the given graph of the circle.
Complete the statement:
Every point on the circle is exactly ▯ units away from the point (▯, ▯).



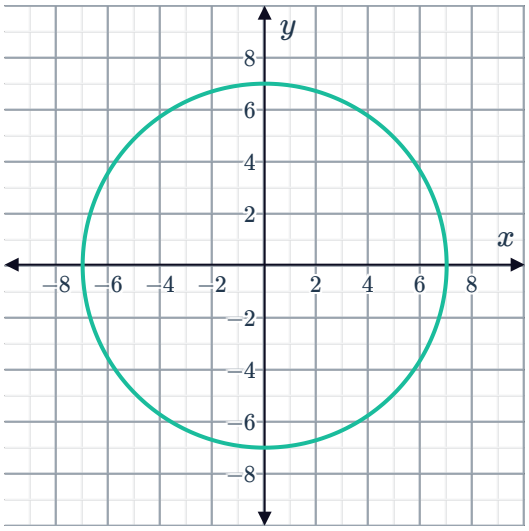
5 How many points lie on the graph of the following equations:

a $x^2 + y^2 = 0$

b $x^2 + y^2 = -81$

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6 State the equation of the following circle:



7 For each of the following circles:

i Sketch the graph of the circle.

ii Write the equation of the circle.

a A circle with center at the origin and a radius of 4 units.

b A circle with center at the origin and a radius of $\sqrt{3}$ units.

c A circle has its center at the origin and a radius of $\sqrt{18}$ units.

8 Given the equation of a circle $x^2 + y^2 = 64$:

a State the coordinates of the center.

b Find the radius.

c Find the x-intercepts.

d Find the y-intercepts.

e Sketch the graph of the circle.

9 The equation of a circle is given by $x^2 + y^2 = 32$. Calculate the exact radius of this circle.

10 For each of the following equation of a circle:

i Find the radius.

ii Sketch the graph of the circle.

a $x^2 + y^2 = 16$

b $x^2 + y^2 = 36$

11 Write down the equation for a circle with center $(0, 0)$ and radius 2.

12 Consider the circle with center $(0, 0)$ that passes through the point $(5, -10)$.

a Find the exact radius of the circle.

b Find the equation of the circle.

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13 A circle with center $(0,0)$ has an x-intercept at $(6,0)$.

a Find the equation of the circle.

b Find the exact area inside the circle.

14 Consider the circle with equation $x^2+y^2=49$.

a Find the diameter of the circle.

b Find the y-values of the points on the circle that have an x-coordinates of 1.

15 Is the point $(-4,2)$ inside, outside or on the circle with equation $x^2+y^2=21$?

16 For each of the following equations:

i Find the center of the circle.

ii Find the radius of the circle.

iii Sketch the graph of the circle.

a $3x^2+3y^2-48=0$

b $(x+4)^2+(y-2)^2=25$

c $\left(x+\frac{1}{2}\right)^2+\left(y+\frac{1}{2}\right)^2=\frac{9}{4}$

d $(x-4)^2+(y+1)^2=36$

e $(x+3)^2+(y+4)^2=16$

f $(x+1)^2+(y+4)^2=25$

g $25x^2+25y^2=400$

17 Sketch the graph of $x^2+y^2=3$.

18 A circle centered at $(7,-8)$ has an x-intercept at $(-10,0)$. Find the exact radius of the circle.

19 For each of the following equations, find:

i The center of the circle.

ii The exact radius of the circle.

a $(x+3.1)^2+\left(y-\frac{3}{2}\right)^2=\frac{9}{16}$

b $(x-6)^2+(y-6)^2=12$

c $(x-0.7)^2+(y+4.4)^2=3$

d $(x-8)^2+(y+1)^2=13$

20 Consider a circle with center at $(3,3)$ and a radius of 6units.

a Sketch the graph.

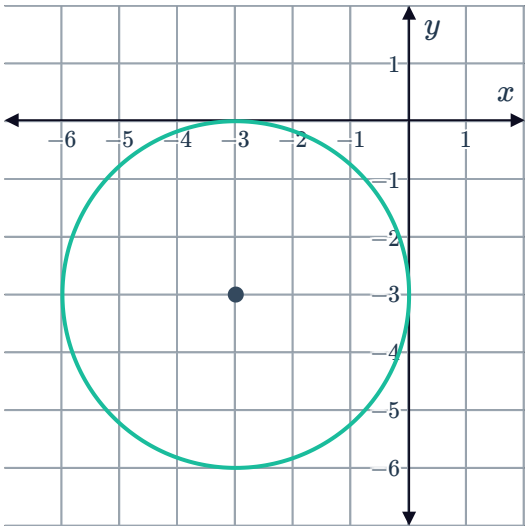
b Find the equation of the circle.

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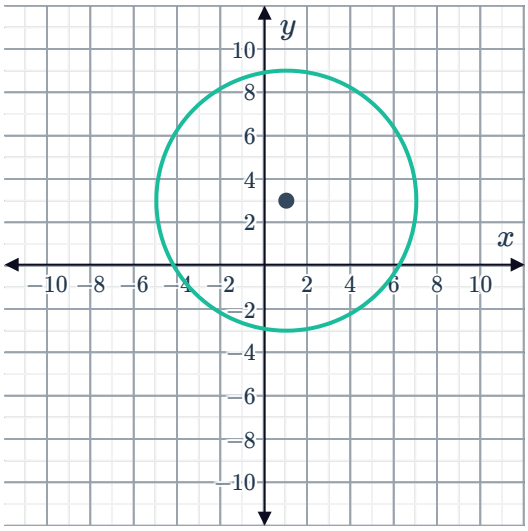
21 For each of the following circles, find:

i The center. **ii** The radius. **iii** The equation.

a



b



22 Write the equation of the circle given the following center and radius:

a Center $(1, -3)$ and radius of 5 units. **b** Center $(-5.7, 0)$ and radius of 3 units.

23 How many x-intercepts does the circle $(x-7)^2 + (y-1)^2 = 4$ have?

24 Consider the circle with center $(8, 6)$ and radius of 8 units.

a Write down the equation of the circle. **b** Does the circle pass through $(0, 0)$?

25 Consider the circle with center $(2, -6)$ and a radius of 3 units.

a Sketch the graph for the given circle.

b Write the equation of the circle in general form.

26 Determine whether the following points lie inside, outside, or on the circle $(x-2)^2 + y^2 = 1$.

a $(1, -1)$ **b** $(3, 0)$ **c** $(3.5, -0.5)$

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27 Determine whether the following points lie inside, outside, or on the circle $(x-2)^2+(y+4)^2=58$.

a $\left(\frac{17}{5},-\frac{6}{5}\right)$ **b** $(-5,-7)$ **c** $(-5,-1)$

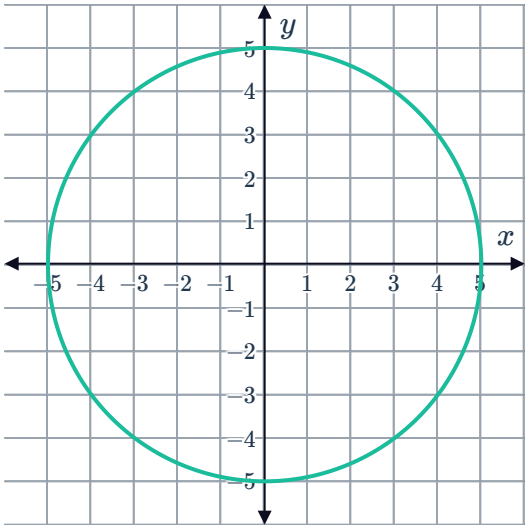
Additional questions

28 State the equation for a circle with center (h,k) and radius r.

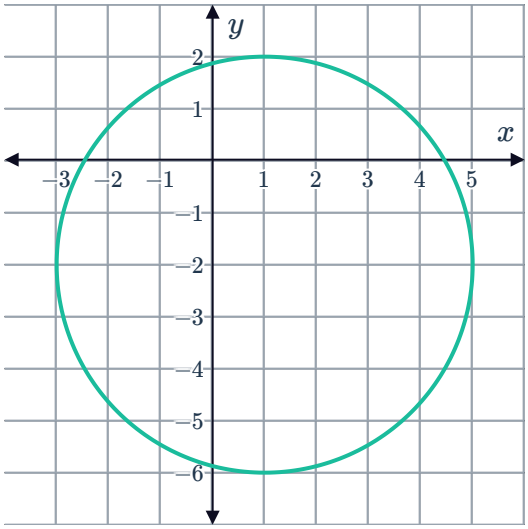
29 For each of the following circles, state:

i The center. **ii** The radius. **iii** The diameter. **iv** The equation.

a



b



30 For a circle with the equation $(x-2)^2+(y+3)^2=75$:

a State the center of the circle **b** Calculate the radius of the circle

c Calculate the diameter of the circle

31 For each of the following descriptions, write the equation of the circle.

a A circle with center at $(0,6)$ and a diameter of 6 units.

b A circle with center at $\left(\frac{17}{2},0\right)$ and a diameter of $2\sqrt{15}$ units.

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32 For each of the following equations:

i Find the center of the circle.

ii Find the radius of the circle.

iii Graph the circle.

iv State the domain and range.

a $x^2 + y^2 = 16$

b $(x+4)^2 + (y-2)^2 = 25$

33 Consider the circle with center $(-2, 5)$ that passes through the point $(3, -7)$.

a Find the radius of the circle

b Find the equation of the circle

34 Determine whether the following points are inside, outside, or on the circle with equation $x^2 + y^2 = 25$?

a $(-3, 2)$ **b** $(4, 3)$ **c** $(1, 6)$ **d** $(-5, 0)$

35 For each of the following equations of a circle:

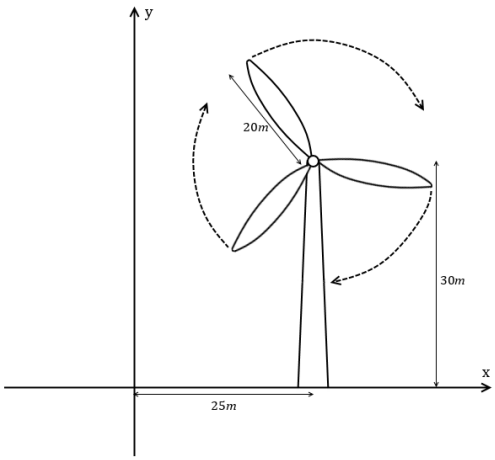
i Rewrite the equation of the circle in standard form by completing the square.

ii Sketch the graph of this circle.

a $x^2 - 12x + y^2 + 2y + 1 = 0$

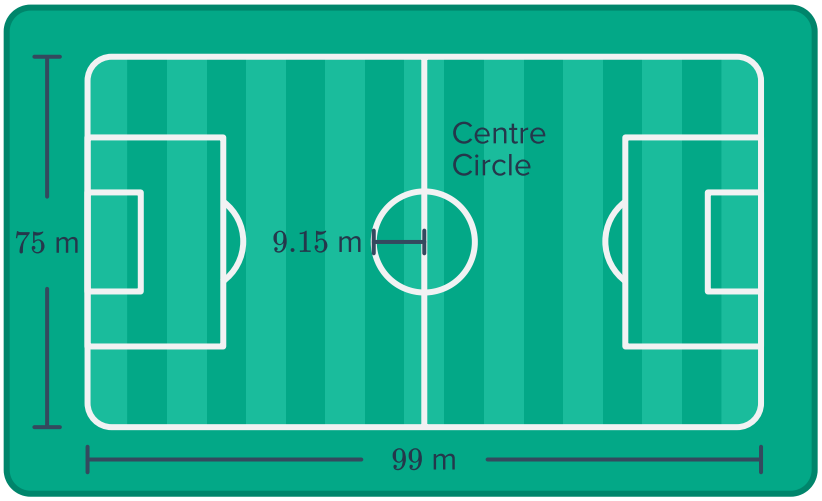
b $4x^2 + 4y^2 + 12x - 36y + 41 = 0$

36 A wind turbine has 20 m long blades that are attached to a tower 30 m high. The distance from the origin to the base of the wind turbine is 25 m. Find the standard form of the equation of the circle represented by the path of the blades of the wind turbine.



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37 A soccer match is being televised.
One of the cameras is mounted on a drone which is programmed to zoom in on the ball only when it is inside the center circle.
The drone uses a coordinate system to track the position of the ball, where the origin is at the bottom left corner of the field and each unit corresponds to $1\ \text{m}$.



a The center of the center circle is in the exact center of the field. What are the coordinates of the circle's center?

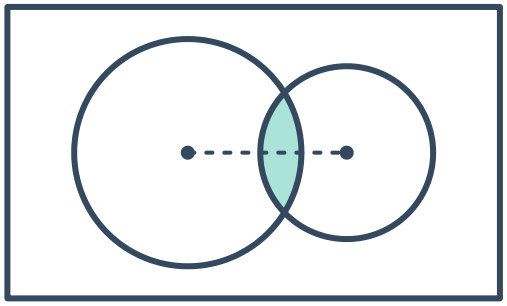
b What is the radius of the center circle?

c State the equation for the circle.

d State the domain and range of the circle in interval notation.

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38 Coralee needs to use her school's laser cutter to make a cork component for her design project.
Using technology to sketch the component, she knows that the intersection of two different sized circles inscribed on a piece of cork gives the shape that she wants.



Two such circles have the equations $(x-12)^2+(y-8)^2=49$ and $(x-20)^2+(y-8)^2=36$, where each unit is 1 cm and $(0,0)$ is the bottom left corner of the given cork board.

a State the domain and range of the leftmost circle in interval notation.

b State the domain and range of the rightmost circle in interval notation.

c What is the width of the component that Coralee wants to make?

d Find the least area of material that could be used to make Coralee's shape, considering the following constraints:

- The laser cutter will only accept rectangular pieces of material.
- The height of the intersection is 10.17 cm.
- There also needs to be a gap of at least 1 cm between any cut and the edge of the piece.

39 A circle has a diameter with endpoints of $A(-1,1)$ and $B(9,-7)$.

a Determine the equation of the circle.

b Prove that the point $P(8,3)$ is outside of the circle.

40 Three receiving stations are located on a coordinate plane, where each unit is one mile, at the points $(-3,-2)$, $(-1,2)$ and $(3,1)$. The epicenter of an earthquake is determined to be 2 miles, 4 miles and 5 miles from these three points respectively.

a Determine the coordinates of the epicenter of the earthquake.

b The earthquake can be felt 10 miles away. Determine if the town located at $(8,-7)$ would feel the earthquake.

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